# Single Oral Practice 2

Module 5 - Single Oral Administration · Daily practice sheet · No date

Name _____ Date _____

A 10-mg single dose of a drug is given orally to a patient. The drug is about 80% absorbed and the equation that fits the observed data is $C_p = 70$ ng/mL $(e^{-0.1t} - e^{-0.3t})$. [Rate constants are in $hr^{-1}$]. Determine the following:

**Part 1** Volume of distribution

**Part 2** Total body clearance

## **Part 3** Half-life of elimination

## **Part 4** Half-life of absorption

## **Part 5** Maximum plasma concentration

**Part 6** Maximum plasma concentration if the dose were increased to 20 mg

**Part 7** Half-life of elimination if the clearance were reduced to 8.57 L/hr, assuming no change in apparent volume of distribution

**Part 8** Maximum plasma concentration for a 10 mg dose if the clearance were reduced to 8.57 L/hr, assuming no change in apparent volume of distribution